Objective Function

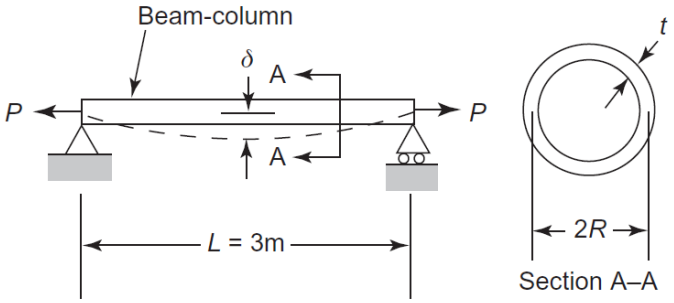
Design Variables

Constraints

CE 587 (STRUCTURAL OPTIMIZATION)
2026-2027 Fall Semester
– ASSIGNMENT 1-
(Due on 28 October 2026)

QUESTION 1.

Design a hollow circular beam-column, shown in the below figure, for two conditions: When P=50 (kN), the axial stress σ must not exceed an allowable value σ_a, and when P=0, deflection δ due to self-weight should satisfy $\delta \leq 0.001L$.



The limits for dimensions are t=0.10 to 1.0 cm, R=2.0 to 20.0 cm, and R/t ≤ 20 (AISC, 2005).

Use the following data: $\delta=5wL^4/384EI$; w=self-weight force/length (N/m); σ_a=250 MPa; modulus of elasticity E=210 GPa; mass density ρ=7800 kg/m^3; $\sigma=P/A$; gravitational constant g=9.80 m/s^2; moment of inertia $I=\pi R^3 t$ (m^4).

a. Formulate the minimum-mass circular beam-column problem.
b. Determine the optimum solution of the problem using graphical method.
c. Formulate and solve the optimum design problem using Microsoft Excel Solver.

QUESTION 2.

A cantilever beam is subjected to the point load P (kN), as shown in Figure. The maximum bending moment in the beam is PL (kN.m) and the maximum shear is P (kN). Formulate the minimum-mass design problem using a hollow circular cross-section.

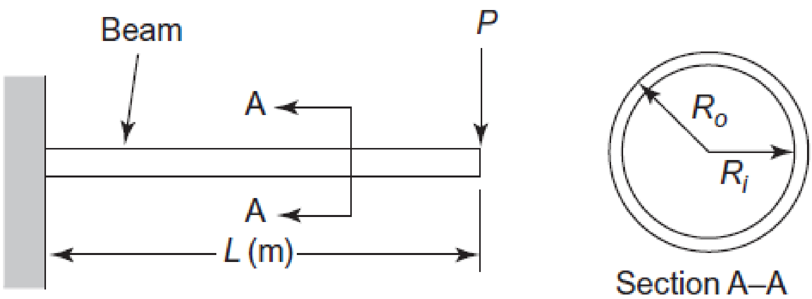
The material should not fail under bending or shear stress. The maximum bending stress is calculated as

$$\sigma = \frac{PL}{I} R_0$$

where $I = \pi(R_0^4 - R_i^4)/4$ is moment of inertia of the cross-section. The maximum shearing stress is calculated as

$$\tau = \frac{P}{3I}(R_0^2 + R_0 R_i + R_i^2)$$

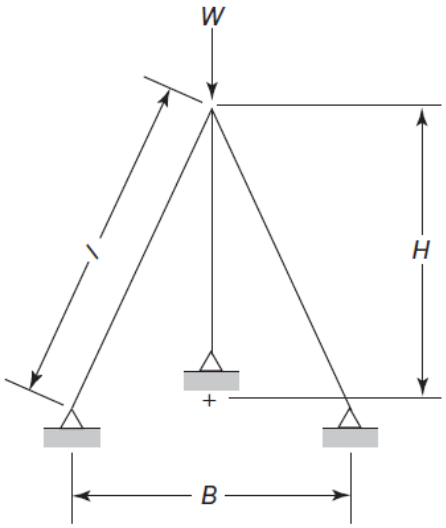
Transcribe the problem into the standard design optimization model (also use $R_o \leq 40.0$ cm, $R_i \leq 40.0$ cm). Use this data: P = 14 kN; L = 10 m; mass density ρ = 7850 kg/m³; allowable bending stress σ_b = 165 MPa; allowable shear stress τ_a = 50 MPa.



d. Formulate the minimum-mass cantilever beam problem.
e. Determine the optimum solution of the problem using graphical method.
f. Formulate and solve the optimum design problem using Microsoft Excel Solver.

QUESTION 3.

Design a minimum mass tripod of height H to support a vertical load W = 60 kN. The tripod base is an equilateral triangle with sides B = 1200 mm. The struts have a solid circular cross-section of diameter D.



The axial stress in the struts must not exceed the allowable stress in compression, and the axial load in the strut P must not exceed the critical buckling load P_{cr} divided by a safety factor FS =2. Use consistent units of Newtons and centimeters. The minimum and maximum values for the design variables are $50 \le H \le 200$ cm and $0.5 \le D \le 10$ cm.

Material properties and other relationships are given next:

Material	aluminum alloy 2014-T6
Allowable compressive stress	$\sigma_a = 150$ MPa
Young's modulus	$E = 75$ GPa
Mass density	$\rho = 2800\ \text{kg/m}^3$
Strut length	$l = \left(H^2 + \frac{1}{3}B^2\right)^{0.5}$
Critical buckling load	$P_{cr} = \frac{\pi^2 EI}{l^2}$
Moment of inertia	$I = \frac{\pi}{64}D^4$
Strut load	$P = \frac{Wl}{3H}$

a. Formulate the minimum-mass tripod design problem.
b. Determine the optimum solution of the problem using graphical method.
c. Formulate and solve the optimum design problem using Microsoft Excel Solver.